

Gabriel Cruz

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EDUCATION

Georgia Institute of Technology

Atlanta, GA

B.S. in Computer Science and Mathematics, GPA: 3.2

Aug 2024 – May 2028

- Threads: **Systems & Architecture**, Intelligence
- Coursework: **Computer Architecture**, **Digital Design Lab**, **Systems & Networks**, Data Structures & Algorithms, Design & Analysis of Algorithms, **Linear Algebra**, Discrete Mathematics, Probability & Statistics

EXPERIENCE

Electrical Engineering Intern

June 2023 – May 2024

Johns Hopkins Applied Physics Laboratory

Laurel, MD

- **Led** small teams building prototypes for **wireless power transfer (WPT)** and **electrical impedance spectroscopy (EIS)**, including a low-cost **2.45 GHz** closed-loop WPT system with **adaptive phase and impedance tuning**
- Researched **microwave signals** for **defense and utility** applications; simulated 2.45 GHz and 5.8 GHz ISM signals in **GNU Radio**, reducing energy rebound by **30%**
- Developed a predictive **Free Space Path Loss** model and **tested existing systems** to improve wireless system design accuracy
- Built an automated leaf health monitoring pipeline using an **Arduino** and Python (Plotly) that **counted xylem** in leaves and flagged whether they fell in a healthy range, achieving **99.1%** detection accuracy

PROJECTS

LC-5200 Microcoded Processor | *CircuitSim, Microcode, Assembly*

- Built a 32-bit microcoded **datapath** (ALU, register file, control unit) and added **interrupt** support, including EI, DI, IN, and RETI instructions and an assembly interrupt handler
- Optimized the processor to a **1.64× average speedup** across three benchmark programs using a **dual-ported register file**, an immediate-to-ALU bypass, and a dedicated **PC incrementer** that cut fetch from 3 cycles to 2
- Found and fixed a **microcode bug** (wrong ALU function in the BEQ compare state) by tracing ALU inputs and outputs

Multithreaded CPU Scheduler | *C, pthreads, Linux*

- Implemented **FCFS**, **Round Robin**, **SRTF**, and **preemptive priority** schedulers for a multiprocessor OS simulator that runs one thread per simulated CPU
- Protected the shared ready queue and per-CPU state with **mutexes** and **condition variables** (blocking idle CPUs), preventing **race conditions** in a **multithreaded** application

Virtual Memory Simulator | *C*

- Implemented **address translation**, **page fault** handling, and swap for a system with a 24-bit virtual address space, 20-bit physical memory, and 16 KB pages
- Built **Clock Sweep (second chance)** and **approximate LRU** page replacement, and computed **AMAT** on benchmark memory traces

Cache-Blocked Matrix Multiplication | *C++*

- Built a custom templated `Matrix<T> C++ class` with **row-major** layout, plus naive and **cache-blocked (tiled)** multiplication
- Benchmarked with a custom **GFLOPs/sec** harness, achieving a **6.3× speedup** (3.0% to 19.1% of theoretical peak) and diagnosing **poor cache reuse** as the bottleneck

TECHNICAL SKILLS

Languages: C++, C, Python, Assembly, **VHDL**, Java

Tools/Concepts: Linux, Git, Make, GDB, **pthreads**, **multithreading**, interrupts, state machines

Hardware: Arduino, FPGA, **Quartus**, RF/Microwave, GNU Radio, Oscilloscope

Libraries: NumPy, SciPy, Pandas, scikit-learn